
MA 705: Modelling Population Dynamics

Course Code: MA 705

Course Name: Modelling Population Dynamics

L-T-P-C: 3-0-0-3

Approval: One-time approval recommended by the 26th BoA and approved by the 18th Senate (earlier listed as an existing discipline elective of the M.Sc. Applied Mathematics programme, 9th Senate)

Course Description

This course is an introduction to synthesis of mathematical models and empirical ecology. Why do organisms become extremely abundant one year, and then apparently disappear a few years later? Why population outbreaks in certain species in certain locations happen more or less regularly, while in other locations such eruptions are irregular? This course will provide answers to such questions. From this course, students will gain a better understanding of population dynamics in general.

Course Content

- **Module 1:** Modeling Nature, General philosophy of the approach.
- **Module 2:** Population Dynamics from the first principle, Exponential Growth, Self Limitation, Consumer Resource Oscillation.
- **Module 3:** Types of Models: Continuous-time, Discrete-time, Delayed Differential models, Exogenous Drivers, Age and Stage- structured Models.
- **Module 4:** Population Interactions: Functional response, Aggregative response, Numerical Response, Lotka Volterra Model, Anatomy of Predator Prey Cycle, Grazing systems, pathogens and parasites, Tritrophic Models.
- **Module 5:** Empirical Approaches: Analysis of Population fluctuations, Time Series Analysis, Fitting Models to Data, Fitting Mechanistic Models.